

Rosen — §9.3: Representing Relations

Selected Exercises

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Exercise 10

How many nonzero entries does the matrix representing the relation R on $A = \{1, 2, 3, \dots, 1000\}$ consisting of the first 1000 positive integers have if R is

- (a) $\{(a, b) \mid a \leq b\}$?
- (b) $\{(a, b) \mid a = b \pm 1\}$?
- (c) $\{(a, b) \mid a + b = 1000\}$?
- (d) $\{(a, b) \mid a + b \leq 1001\}$?
- (e) $\{(a, b) \mid a \neq 0\}$?

Exercise 14

Let R_1 and R_2 be relations on a set A represented by the matrices

$$M_{R_1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad M_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Find the matrices that represent

- (a) $R_1 \cup R_2$.
- (b) $R_1 \cap R_2$.
- (c) $R_2 \circ R_1$.
- (d) $R_1 \circ R_1$.
- (e) $R_1 \oplus R_2$.

Exercise 15

Let R be the relation represented by the matrix

$$M_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Find the matrices that represent

(a) R^2 .

(b) R^3 .

(c) R^4 .

Exercise 17

Let R be a relation on a set A with n elements. If there are k nonzero entries in M_R , the matrix representing R , how many nonzero entries are there in $M_{\overline{R}}$, the matrix representing $\overline{R}$, the complement of R ?

Exercise 29

How can the directed graph of a relation R on a finite set A be used to determine whether a relation is asymmetric?

Exercise 35

Show that if M_R is the matrix representing the relation R , then $M_R^{[n]}$ is the matrix representing the relation R^n .

Exercise 36

Given the directed graphs representing two relations, how can the directed graph of the union, intersection, symmetric difference, difference, and composition of these relations be found?